

Building Secure AI Agents with OpenClaw on AWS

Published by [O'Reilly Media, Inc.](#)



Deploy, Customize, and Harden AI Agents on AWS

Complete this course and earn a badge! 

[Is this live event for you?](#) [Schedule](#)

Course Outcomes

- Deploy OpenClaw on Amazon Lightsail with Amazon Bedrock, configuring the instance, pairing your browser, enabling AI capabilities, and connecting a messaging channel
- Build custom OpenClaw skills using the SKILL.md format, including tool declarations, script integration, and ClawHub publishing
- Design persistent memory systems using SOUL.md, HEARTBEAT.md, and the memory folder to create agents that learn and adapt across sessions
- Implement security hardening using IAM least-privilege policies, Docker sandboxing, gateway token rotation, and third-party skill auditing to protect against prompt injection, data exfiltration, and malicious skills

April 27
10am-2pm Eastern Daylight Time
154 Spots Remaining

[Sign up!](#)

Your Instructor



Kesha Williams

Kesha Williams is an enterprise architect and AI consultant with over three decades of experience designing, building, and operating production software systems. She's the founder and managing partner of Keysoft,...

[Read more](#)

[in](#)
[X](#)
[e](#)
[Q](#)

Associated roles

[AI engineer](#)
[Android developer](#)

[Backend developer](#)
[Business analyst](#)

[+28 more](#)

Skills covered

[AI Agents](#)
[Generative AI](#)

MODULE 1: OpenClaw Architecture & AWS Deployment (10:00 - 10:35)

Topics: The "brain, hands, memory" mental model. Gateway architecture. How OpenClaw connects to LLMs via Amazon Bedrock. Why the Lightsail blueprint changes the deployment story.

Hands-on: Launch an OpenClaw instance on Lightsail. SSH in, copy the gateway token, and pair your browser. Run the CloudShell script to enable Bedrock. Send your first message.

Break: 10:30 - 10:35

MODULE 2: Configuration & Messaging Integration (10:35 - 11:15)

Topics: SOUL.md and agent identity. Built-in tools (exec, read, write, edit, browser, web_search, web_fetch, message, cron) and how they're controlled via openclaw.json. SKILL.md anatomy. Tools vs. skills and why the distinction matters for security. Connecting messaging platforms. Configuring exec settings for skills development.

Hands-on: Write a SOUL.md that defines your agent's personality, boundaries, and behavioral rules. Configure tools.allow in openclaw.json. Connect your agent to Telegram and have a conversation from your phone. Configure exec settings for the skills exercises in Module 3.

Break: 11:10 - 11:15

MODULE 3: Building Custom Skills (11:15 - 12:15)

Topics: Skill architecture (SKILL.md frontmatter, body instructions, optional scripts). Three-layer loading system. Skill precedence. How skills reference tools. Publishing to ClawHub.

Hands-on: Build three custom skills from scratch on your Lightsail instance.

- Skill 1 — Project Scaffolder: Creates a project directory structure with starter files from a description of the project type and name
- Skill 2 — File Organizer: Sorts files in a directory into categorized subdirectories by type and date
- Skill 3 — Code Review: Analyzes code and provides structured feedback with security, testing, complexity, and style categories

Test each skill via natural language. Observe how the agent selects and loads skills at runtime.

Break: 12:10 - 12:15

MODULE 4: Sessions, Memory & Proactive Behavior (12:15 - 1:00)

Topics: Session architecture and how conversations are tracked. Compaction and what happens when context fills up. Memory system: MEMORY.md, topic-based memory notes, memory search. HEARTBEAT.md and scheduled tasks. Proactive agents. Persistence on Lightsail.

Hands-on: Explore session state and memory files on disk. Tell the agent something memorable and watch it write to memory. Set up HEARTBEAT.md with a scheduled monitoring task. Restart the agent and verify it retains context across restarts.

Break: 12:55 - 1:00

MODULE 5: Security Hardening on AWS (1:00 - 1:45)

Topics: The three-layer threat model (infrastructure, application, configuration). Attack chains: Server Takeover, Persistent Manipulation, Invisible Backdoor. Bedrock Guardrails. IAM least-privilege. Exec sandboxing. Security audit. OWASP Top 10 for Agentic Applications 2026.

Hands-on exercises:

- Exercise 1: Patch the OS, restrict the Lightsail firewall to your IP, run openclaw security audit --fix
- Exercise 2: Remove Marketplace permissions from the IAM policy, add bedrock:ApplyGuardrail
- Exercise 3: Create a Bedrock Guardrail with a denied topic in the Bedrock console (same region as your instance), wire it to OpenClaw, and test
- Exercise 4: Walk through the 15-item production security checklist

WRAP-UP & Q&A (1:45 - 2:00)

What you built today. Recommended next steps. Resources and follow-up. Final Q&A.

Resources

- [Course GitHub Repository](#): starter configs, skill templates, IAM policy templates, security checklists
- [OpenClaw Documentation](#)
- [OpenClaw Security Docs](#)
- [AWS Lightsail OpenClaw Quick Start](#)
- [OWASP Top 10 for Agentic Applications 2026](#)